B01+B02 文献管理和报告撰写

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2025 年 9 月 26 日

我已掌握 latex 较为基础的功能，并曾使用它书写过一些课程作业和完成数模竞赛，我将会贴出我用 latex 书写的一次作业，并补全作业中我未使用的要求的贴图片和引用文献的功能。

1　表格制作示例

表 1 展示了一个简单的数据表格示例。

表 1: 示例表格

编号	名称	数值	备注
1	数据 A	10.5	正常
2	数据 B	25.3	偏高
3	数据 C	8.7	偏低

2　文献引用示例

以下是文献引用的示例：
引用两个文献[1;2] 的方式

3　结论

综上所述，本文展示了 LaTeX 中语句书写、公式输入、图片插入、表格制作和文献引用的基本方法。

参考文献

[1] Author A. (2020). 论文标题. 期刊名称, 卷 (期), 页码.

[2] Author A., Author B., & Author C. (2021). 另一篇论文. 会议名称, 页码.

```latex
\documentclass{article}
\usepackage{amsmath}
\usepackage[UTF8]{ctex}
\usepackage{enumitem}
%opening
\title{\textbf{Atomic Physics\\The first homework}}
\author{24349020 \quad 周文斌}

\begin{document}

\maketitle

\section*{1-1}
\noindent
nswer:

According to the relation between collision coefficient b and scattering
angle $\theta$
\begin{equation}
    b=\frac{a}{2}\cot{\frac{\theta}{2}}
\end{equation}
The smaller the collision coefficient b,the larger the scattering angle.So
the scattering angle is maximized when the $\alpha$ particle collides
centrally with the electron.

Due to the elastic collisions,according to the conservation of momentum and
energy:
\begin{equation}
\begin{aligned}
    \frac{1}{2}Mv^2 &= \frac{1}{2}Mv_1^2 + \frac{1}{2}mv_2^2\\
    M\vec{v} &= M\vec{v}_1 + m\vec{v}_2\\
    m &\ll M
\end{aligned}
\end{equation}
Solution:\begin{equation*}
    \begin{aligned}
        v_2&=2v\\v_1&\approx v
    \end{aligned}
\end{equation*}
So,we can get:
```

1 列: 0 插入

图 1:

```latex
documentclass[]{article}
usepackage{amsmath}
usepackage{geometry}
usepackage{ctex}
usepackage{graphicx}
geometry{a4paper, left=3.5cm, right=3.5cm, top=2cm, bottom=3cm}
title{\textbf{B01+B02 文献管理和报告撰写}}
author{24349020\quad Wenbin Zhou}

begin{document}
aketitle
已掌握latex较为基础的功能，并曾使用它书写过一些课程作业和完成数模竞赛，我将会贴出我
latex书写的一次作业，并补全作业中我未使用的要求的贴图片和引用文献的功能。\\
wo picture:
begin{figure}
    \includegraphics[width=0.3\linewidth]{D:/TeX_files/GPE/screenshot001}
    \caption{}
    \label{fig:screenshot001}
end{figure}

end{document}
```

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文献的功能。
Two picture:

图 2:

Theoretical Mechanics
Chapter 1

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September 18, 2025

1.2

Let the angle between the tangent to a point on the cycloid and the horizontal direction be φ.
From the equations provided by the question:

$$\begin{cases} x = a(\theta - \sin\theta) \\ y = -a(1 - \cos\theta) \end{cases}$$

We can get:

$$\tan\varphi = \frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{-\sin\theta}{1-\cos\theta} = -\frac{1}{\tan\frac{\theta}{2}} \tag{1}$$

$$\begin{cases} \ddot{x} = a(\ddot{\theta} - \ddot{\theta}\cos\theta + \dot{\theta}^2\sin\theta) \\ \ddot{y} = -a(\ddot{\theta}\sin\theta + \dot{\theta}^2\cos\theta) \end{cases} \tag{2}$$

According to Newton's second law, we give the dynamic equations in horizontal and vertical directions:

$$\begin{cases} m\ddot{x} = -N\sin\varphi \\ m\ddot{y} = -mg + N\cos\varphi \end{cases} \tag{3}$$

So we can get:

$$\tan\varphi = \frac{-\ddot{x}}{\ddot{y}+g} \tag{4}$$

Union(1)(2)(4):

$$2a\ddot{\theta}\sin\frac{\theta}{2} + a\dot{\theta}^2\cos\frac{\theta}{2} - g\cos\frac{\theta}{2} = 0 \tag{5}$$

Define:$\xi = \cos\frac{\theta}{2}$,then

$$\dot{\xi} = -\frac{\dot{\theta}}{2}\sin\frac{\theta}{2}$$

$$\ddot{\xi} = -\frac{\ddot{\theta}}{2}\sin\frac{\theta}{2} - \frac{\dot{\theta}^2}{4}\cos\frac{\theta}{2}$$

Substitute into the differential equation and rearrange to obtain:

$$4a\ddot{\xi} + g\xi = 0$$

Solve this equation,we can get:

$$\xi = A_\xi\cos(\omega t + \phi) \tag{6}$$

$\omega = \sqrt{\frac{g}{4a}}$is the angular frequency $\phi$is the initial phase $A_\xi$is the amplitude of ξ

$$\theta = 2\arccos[A_\xi\cos(\omega t + \phi)] \tag{7}$$

$$T = \frac{2\pi}{\omega} = 4\pi\sqrt{\frac{a}{g}} \tag{8}$$

is the period

$$A_\theta = 2\arccos A_\xi \tag{9}$$

is the amplitude of θ

Apparently,period is independent with amplitude

1.12

Let the angle between the tangent line and the x-axis be θ.

$$\vec{a} = \frac{dv}{dt}\vec{e_t} + \frac{v^2}{\rho}\vec{e_n} \tag{10}$$

From the question,we see that

$$\frac{dv}{dt} = a\frac{v^2}{\rho} \tag{11}$$

By the chain rule,we have:

$$\frac{dv}{dt} = \frac{dv}{ds}\frac{ds}{dt} = v\frac{dv}{ds} = a\frac{v^2}{\rho}$$

By transformation,we obtain:

$$\frac{dv}{v} = a\frac{v}{\rho} = a\,d\theta$$

Integrate:

$$\int_u^v \frac{dv}{v} = \int_{\theta_A}^{\theta} a d\theta$$

$$\ln\frac{v}{u} = a(\theta - \theta_A)$$

$$v = u\,e^{a(\theta-\theta_A)} \tag{12}$$

At the origin

$$v = v_0, \theta = \theta_0 = \frac{\pi}{2}$$

At A point

$$\theta_A = \arctan\frac{dy}{dx} = \arctan\frac{p}{y} = \arctan 1 = \frac{\pi}{4}$$

Thus,we can get:

$$v_0 = u\,e^{a\frac{\pi}{4}}$$

It is easy to see that θ jumps discontinuously from $\frac{\pi}{2}$ to$-\frac{\pi}{2}$ at the origin.
At B point

$$\theta_B = \arctan\frac{p}{y} = \arctan -1 = -\frac{\pi}{4}$$

So:

$$v_B = v_0\,e^{a(\theta_B - \theta_0')} = u\,e^{a\frac{\pi}{2}} \tag{13}$$

1.13

Answer:

It is easy to see that the net external force acting on the system of particles is $m_B g$. Therefore,the momentum of the system is not conserved.

The net external torque about point O is 0,so the angular momentum of the system is conserved.

The net external force $m_B g$ is conservative force,so the energy of the system is conserved.
The dynamic equations for A are:

$$\begin{cases} m_A(\ddot{r} - r\dot{\theta}^2) = -T \\ m_A(r\ddot{\theta} + 2\dot{r}\dot{\theta}) = 0 \end{cases} \tag{14}$$

The dynamic equation for B is:

$$m_B\ddot{h} = m_B g - T \tag{15}$$

The constraint equation is:

$$r + h = l \tag{16}$$

According to the second equation in (14) and the initial conditions $r\big|_{t=0} = a, r\dot{\theta}\big|_{t=0} = v_0$

$$r\frac{d\dot{\theta}}{dt} = -2\frac{dr}{dt}\dot{\theta}$$

$$\frac{d\dot{\theta}}{\theta} = -2\frac{dr}{r}$$

Integrate:

$$\dot{\theta} = \frac{\dot{\theta}\big|_{t=0}(r\big|_{t=0})^2}{r^2} = \frac{av_0}{r^2} \tag{17}$$

Using chain rule:

$$\ddot{r} = \frac{d\dot{r}}{dt} = \frac{d\dot{r}}{dr}\frac{dr}{dt} = \dot{r}\frac{d\dot{r}}{r} \tag{18}$$

Union(14)(15)(16)and eliminate h and T using the condition $m_A = m_B = m$

$$2\ddot{r} + g = r\dot{\theta}^2 \tag{19}$$

Substitute (17)(18) into and rearrange:

$$2\dot{r}d\dot{r} = (\frac{a^2v_0^2}{r^3} - g)dr$$

Integrate and substitute $v_0 = \sqrt{\frac{9ag}{2}}$ into:

$$\dot{r}^2 = -\frac{9a^3g}{4r^2} - gr + \frac{13ag}{4} \tag{20}$$

At the closest and farthest points, $\dot{r}^2 = 0$ i.e.:

$$0 = -\frac{9a^3g}{4r^2} - gr + \frac{13ag}{4} = -\frac{g}{4r^2}(r-a)(r-3a)(4r+3a) \tag{21}$$

So

$$a \le r \le 3a$$

1.19

$$L = r \cdot \rho\pi r \cdot \dot{\theta}r = \rho\pi r^3\dot{\theta}$$

$$\dot{L} = \rho\pi r^3\ddot{\theta} \tag{22}$$

$$dM_1 = r\cos\theta dmg = r^2\rho g\cos\theta d\theta$$

Integrate:

$$M_1 = \int_0^{\pi-\theta} = r^2\rho g\cos\theta d\theta = r^2\rho g\sin\theta$$

$$M_2 = r \cdot \rho\theta rg = \rho r^2\theta g$$

$$M = M_1 + M_2 = \rho r^2 g(\theta + \sin\theta) \tag{23}$$

According to angular momentum theorem:

$$\dot{L} = M$$

i.e.

$$\rho\pi r^3\ddot{\theta} = \rho r^2 g(\theta + \sin\theta)$$

Rearrange:

$$\ddot{\theta} = \frac{g}{r\pi}(\theta + \sin\theta) \tag{24}$$

Using chain rule:

$$\ddot{\theta} = \frac{\dot{d\theta}}{dt} = \frac{d\dot{\theta}}{d\theta}\frac{d\theta}{dt} = \dot{\theta}\frac{d\dot{\theta}}{d\theta}$$

It is easy to see the initial conditions $\theta\big|_{t=0} = 0, \dot{\theta}\big|_{t=0} = 0$.
Integrate:

$$\dot{\theta} = \sqrt{\frac{g}{r\pi}(\theta^2 + 4\sin^2\frac{\theta}{2})} \tag{25}$$

So

$$v = r\dot{\theta} = \sqrt{\frac{gr}{\pi}(\theta^2 + 4\sin^2\frac{\theta}{2})} \tag{26}$$

1.24

Equation of ring for the origin
$O' \to O$:

$$\vec{r_{o'}} = R\cos\omega t\vec{e_x} + R\sin\omega t\vec{e_y}$$

$m \to O'$

$$\vec{r'} = R\cos\theta'\vec{e_x} + R\sin\theta'\vec{e_y}$$

So,$m \to O$:

$$\vec{r} = \vec{r'} + \vec{r_{o'}} = R(\cos\omega t + \cos\theta')\vec{e_x} + R(\sin\omega t + \sin\theta')\vec{e_y}$$

$$\ddot{\vec{r}} = -R(\omega^2\cos\omega t + \ddot{\theta}'\sin\theta' + \dot{\theta'}^2\cos\theta')\vec{e_x} + R(\ddot{\theta}' - \dot{\theta'}^2\sin\theta' - \omega^2\sin\omega t)\vec{e_y} \tag{27}$$

According to Newton second law:

$$\vec{F} = m\ddot{\vec{r}}$$

We get:

$$\begin{cases} F\cos\theta' = -mR(\omega^2\cos\omega t + \ddot{\theta}'\sin\theta' + \dot{\theta'}^2\cos\theta') \\ F\sin\theta' = mR(\ddot{\theta}'\cos\theta' - \dot{\theta'}^2\sin\theta' - \omega^2\sin\omega t) \end{cases} \tag{28}$$

Union and rearrange:

$$\ddot{\theta}' + \omega^2\sin(\theta' - \omega t) = 0 \tag{29}$$

By the geometric relation$2\theta = \theta' - \omega t$,we get:

$$\ddot{\theta}' = 2\ddot{\theta}$$

Substitute into (29)

$$\ddot{\theta} + \omega^2\sin\theta\cos\theta \tag{30}$$

1.28

Because the force on the ball is always perpendicular to its direction of motion, so the energy is conserved.
The mass point is subject to a centripetal force directed along the tangent to the circle,so the torque about the axis through the center isn't 0,the angular momentum about the axis is not conserved. Consider the state of the system at an instant after the ball started moving.Let the tangent point at this instant be O':

$$O'(R\cos\theta, R\sin\theta)$$

$m \to O'$:

$$x' = R\theta\sin\theta \quad y' = -R\theta\cos\theta$$

$m \to O$:

$$x = R(\cos\theta + \theta\sin\theta) \quad y = R(\sin\theta - \theta\cos\theta)$$

So,we get:

$$\begin{cases} \ddot{x} = \dot{\theta}^2\cos\theta + \theta\ddot{\theta}\cos\theta - \theta\dot{\theta}^2\sin\theta \\ \ddot{y} = \dot{\theta}^2\sin\theta + \theta\ddot{\theta}\cos\theta + \theta\dot{\theta}^2\cos\theta \end{cases} \tag{31}$$

According to the Newton second law:

$$\begin{cases} m\ddot{x} = -T\sin\theta \\ m\ddot{y} = T\cos\theta \end{cases} \tag{32}$$

Substitute (31) into and rearrange:

$$\dot{\theta}^2 + \theta\ddot{\theta} = \frac{d(\theta\dot{\theta})}{dt} = 0 \tag{33}$$

$$T = mR(\dot{\theta}^4 + \dot{\theta}^4\theta^2 + 2\theta\ddot{\theta}\dot{\theta}^2 + \theta^2\ddot{\theta}^2)^{1/2} \tag{34}$$

According to (31),

$$\theta\dot{\theta} = C$$

Because $v\big|_{t=0} = v_0$,we get

$$C = \frac{v_0}{R}$$

Integrate(33),and $\theta\big|_{t=0} = 0$,we get

$$\theta = \sqrt{\frac{2v_0 t}{R}} \tag{35}$$

Substitute into(34),we get

$$T = mv_0\sqrt{\frac{v_0}{2Rt}} \tag{36}$$

1.30

According to Newton second law:

$$\begin{cases} -mR\dot{\theta}^2 = mg\sin\theta - N \\ mR\ddot{\theta} = mg\cos\theta - \mu N \end{cases} \tag{37}$$

Differentiate both sides of the first equation with respect to θ

$$\frac{d(mR\dot{\theta}^2)}{d\theta} = \frac{dN}{d\theta} - mg\cos\theta$$

Using the chain rule

$$\ddot{\theta} = \frac{d\dot{\theta}}{dt} = \frac{d\dot{\theta}}{d\theta}\frac{d\theta}{dt} = \dot{\theta}\frac{d\dot{\theta}}{d\theta} = \frac{1}{2}\frac{d\dot{\theta}^2}{d\theta}$$

So rearrange the former equation:

$$2mR\ddot{\theta} = \frac{dN}{d\theta} - mg\cos\theta \tag{38}$$

Union (38) and the second equation in (37)

$$\frac{dN}{d\theta} + 2\mu N - 3mg\cos\theta = 0 \tag{39}$$

Using the initial condition $N\big|_{\theta=0} = 0$,solve the differential equation:

$$N = -\frac{6\mu mg}{1+4\mu^2}e^{-2\mu\theta} + \frac{3mg(2\mu\cos\theta + \sin\theta)}{1+4\mu^2} \tag{40}$$

Using $N\big|_{\theta=\pi/2} = mg$,we can get:

$$mg = -\frac{6\mu mg}{1+4\mu^2}e^{-\mu\pi} + \frac{3mg}{1+4\mu^2}$$

Rearrange:

$$3\mu e^{-\mu\pi} = 1 - 2\mu^2 \tag{41}$$